

BLOCKER II ROOT CAUSE REFUTED (2026-07-13). Blocker I (OIFS coastline crash) stands. **Blocker II's stated root cause — “a geostrophically-unbalanced remapped state”, cured by a thermal-wind velocity init — is FALSIFIED.** That fix was implemented and failed its own offline gate. **A cold start on the very same PISM cavity, at the same 1200 s timestep, is stable**, so neither the geometry nor the balance argument is the problem; the killer is the *dynamics seam* between the bit-exactly preserved evolved state and the patched changed nodes. Read `report/dynamics_seam_2026-07-13`, then `report/increment_test_2026-07-13`. The evolution/step-1 analysis and figures below remain valid as *observations*; the *causal claim* does not.

Two Coastal-Margin Blockers in the AWI-ESM3 ↔ PISM Moving-Cavity Coupling

A measurement-driven investigation log

- I.** OIFS step-0 coastline crash (*resolved*) • **II.** FESOM cavity-margin vertical-CFL instability
(*root cause: a geostrophically-unbalanced remapped state — restart-value fixes tested & excluded*
— §1, read first)

Experiment `movcav4`, chunk-3 (`awiesm3` 1901)

2026-07-05

Abstract

We are building an end-to-end AWI-ESM3 (FESOM2 + OIFS-48r1 + LPJ-GUESS) ↔ Antarctic-PISM *moving-cavity* coupled run driven by `esm-tools`. On each mesh-change leg the coupling machinery regenerates the FESOM sub-mesh, the OASIS grids/masks/weights, and the OIFS ICMGG initial file so all components agree on the new coastline. The first mesh-change leg (`movcav4`, `awiesm3` 1901) surfaced **two distinct blockers, both localised to the Antarctic coastal margin**. This log records the investigation as a sequence of *measurements* rather than hypotheses.

Blocker I — an OIFS step-0 surface crash — is SOLVED. A floating-point NaN in the surface/moist-physics column struck exactly the coastline cells whose land-sea type flipped. Root cause: the coupling tool `ocp-tool` flipped a cell by setting *only* its land-sea mask and soil type, leaving every other surface field at the old type (a “half-converted” column); OIFS faithfully re-ingested that inconsistency from the regenerated ICMGG and NaN’d in the moist physics. Fix: a masked nearest-neighbour rebuild of each flipped cell in `ocp-tool`, plus the ICMGG re-ingest on warm start. A residual crash then proved *DDH-diagnostic-only*; with DDH off, OIFS runs ~700 coupling steps under `-fpe0`, so OIFS and the fluxes it sends the ocean are finite. Part II (§5 ff.) is the full record.

Blocker II — a FESOM vertical-CFL blowup — is the LIVE blocker, now root-caused. Some 5–10 model days in, FESOM blows up with a vertical-CFL runaway ringing the cavity margin. The cause is a **geostrophically-unbalanced remapped state**: the mesh change plants fresh cavity fronts against salty open water but *zeroes* the balancing current, so the model’s $t = 0$ adjustment grows w at the fronts. This is the *same* instability the hourly movie showed *growing over days* — a $t = 0$ imbalance that amplifies, not a melt process from a balanced start. Every restart-*value* fix (freezing cap; coherent T/S refill to zero density inversions) was tested and does **not** prevent it; a 60s step clears the *fast* mode but a *slow* barotropic `eta_n` mode then grows. Fix: initialise the changed-cell velocity in balance. **§1 is the current truth — read it first.**

1 Current state (2026-07-06, rev. 3): the blowup is a *dynamical imbalance* of the remapped state, not a restart-value pathology

Read this first — supersedes §2 (rev. 2). The rev. 2 basal-melt / thin-layer story led to two concrete restart fixes — a freezing-point cap, then a full coherent-column T/S refill. Both are now **implemented and tested** on a clean leg (`movcav8`, `awiesm3` 1901); together they make the remapped restart **provably clean** (zero static density inversions anywhere) — yet **neither prevents the blowup**. The true root is that the remap hands FESOM a state that is **out of geostrophic balance at the moving margins**: real (and fill-sharpened) lateral density fronts carried across the mesh change, with the balancing current **zeroed out**. That single imbalance expresses *fast* as the vertical-CFL runaway (1200s) and *slow* as a barotropic `eta_n` runaway (60s).

1.1 What was tested since rev. 2, and what it falsified

1. **Freezing-point cap — applied, no effect.** `cap_new_cavity_temp` ran in `movcav8`'s `couple.in` (1968 nodes / 14 670 cells capped to T_f). The leg still blew (`mstep`≈205, `CFLz`≈5.7). Removing the thermal driving does *not* stop it.
2. **Coherent-column T/S refill — all inversions removed, still blows.** The per-level unmasked nearest-node fill (which planted melt-fresh $S \approx 22$ mid-column) was replaced by (a) a *mask-aware open-ocean donor* for surface-opened columns and (b) an *own shelf-base hold* for still-sub-ice retreated columns (committed, `ad4e84b2`). Over all 1595 retreated cells the worst static stability improved $-8.8 \rightarrow -0.5 \text{ kg m}^{-3}$ (11 severe $\rightarrow$ 0). The leg *still* blew (`mstep`≈164, `CFLz`≈8.5), essentially unchanged. **Restart $T/S/\text{ssh}/\text{density}$ is decisively excluded** (four inversion-free variants all blew the same way).
3. **Timestep = 60 s — clears the fast blowup, exposes a slow one.** With `step_per_day`=1440 (everything else identical) the run made 0 `CFLz` warnings and reached `mstep`= 14 629 (≈10 model-days, $4\times$ longer), then died on `eta_n`→NaN (barotropic ssh) at 112.8/−65.9. So the *fast* blowup is genuinely vertical-CFL (timestep-curable) but a *slow barotropic* mode remains — the Δt ramp alone is **not** a cure.

1.2 The initiation mechanism (fast blowup): unbalanced fronts + zero velocity

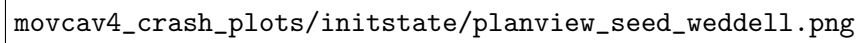
The updated plots (`movcav8` v4 restart; PolyCollection on the real mesh triangles, dateline-wrapping triangles filtered, layer-index vertical axis so the thin cavity layers are visible; `movcav4_crash_plots/initstate/`, Figs. 1, 2) show it directly. Around the *earliest* `CFLz` seed (Weddell, −62.96/−66.81, first warning `mstep` 4, peak level 11 ≈ 95 m):

- **A picket-fence of fresh cavity columns.** Opened cells (ice retreated) come out **uniformly fresh**, $S \approx 31$, punched into an $S \approx 33$ –34 background — strong **lateral density fronts at the grid scale** (≈25 km), strongest mid-column. Advanced cells (open→new ice) keep salty ≈34 water and are comparatively benign — so the *opened* cells are the troublemakers.
- **Velocity is exactly zero across the whole opened region** — the remap zeroes u/v on every changed element; the speed panels are solid black except the far-field unchanged edge.
- A density front *with zero velocity* is **not in thermal-wind balance**: at $t = 0$ the model must spin up the missing geostrophic current, and its **ageostrophic part is w** . The w panels show exactly that — a down/upwelling dipole **pinned to the fronts**, mid-deep layers. At $\Delta t = 1200 \text{ s}$ over ≈10 m layers this w (≈2 mm s^{−1}) gives `CFLz`> 1 **within** ≈4–10 steps.

This is precisely why the T/S *stability* fixes did nothing: the imbalance is **lateral (fronts) + missing momentum**, not a vertical inversion. The *over-freshening* is partly self-inflicted — the shelf-base “hold upward” plants the freshest (ice-base) water throughout the newly-opened column, sharpening the very fronts that drive the adjustment.

1.3 Peak `CFLz` is a consequence, not the seed

The *highest-CFLz* cell at the crash (90.5/−63.5, level 38, **3775 m**, `CFLz` 8.45) is **deep, fully-unchanged open ocean with no cavity at all**, and a perfectly smooth initial state (S varies 0.02 PSU; Fig. 2, top). It is a *late* manifestation: the instability seeds at the opened margins and **advects/propagates into the deep basin by the crash step**. So “high `CFLz` at the crash” ≠ “where it started” — only the earliest (margin) cells show the initiating imbalance.



movcav4_crash_plots/initstate/planview_seed_weddell.png



movcav4_crash_plots/initstate/section_lat_seed_weddell.png

1.4 The slow (60 s) blowup is the same root, a different mode

At 60 s the vertical CFL stays subcritical, so instead a **barotropic η_n response accumulates over ~ 10 days** and piles against the coast/slope *adjacent* to the changed margins. The 60 s crash site (112.8/−65.9, Fig. 2, bottom) is a well-stratified continental slope (warm CDW at depth, surface jet) — a *benign* local state, but only 32 km from the advanced-ice margin (9 changed cells within 100 km). So $\eta_n \rightarrow \text{NaN}$ there is a **slow remote pile-up**, not a local shock. Fast (CFLz) and slow (η_n) are **two faces of one imbalance**.

1.5 Reconciling the two views: a $t = 0$ imbalance that *amplifies*

The rev. 2 hourly movie (movcav4; Figs. 3, 4) and this step-1 analysis (movcav8) are the *same* instability at two times; they reconcile once “clean” is read correctly:

- **“Hour 1 clean” meant clean *values*, not a balanced state.** $T/S/\text{ssh}$ are in normal ranges with no NaNs, but the state is out of geostrophic balance (fresh margin fronts vs salty open water, $u/v=0$). The movie’s “ $w \approx 0$ at hour 1” is that zero initial velocity.
- **The day-scale “growth” is the adjustment of the $t = 0$ imbalance**, not a new process — the missing geostrophic current spins up, w amplifies at the fronts, the fresh/cold band saturates and spreads: exactly the movie’s onset→runaway.
- **Melt is a secondary amplifier, not the driver.** rev. 2 pinned the freshening on melt under new ice (*advanced* cells); the fresh fronts are largely a remap-fill feature at the *opened* cells, present at $t = 0$, and the freezing cap (which removes the melt driving) does not stop the blowup.

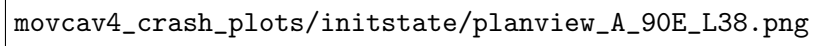
So rev. 2 read the *character* right (a fresh-fronted ~ 95 m coastal baroclinic mode that grows over days) but mis-read normal restart *values* as a balanced IC and over-credited melt. The unifying cause is the unbalanced remapped state.

1.6 Root cause and ranked fix

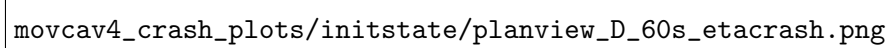
On the spun-up full mesh these fronts carry their balancing geostrophic currents; the remap preserves T/S (density) but **discards the momentum field on changed cells (zeroes u/v) and over-sharpens the opened columns**, handing the model a state far from balance. It must then adjust — fast (vertical CFL) or slow (barotropic). Ranked:

1. **Initialise u/v in thermal-wind balance with the remapped density** (or copy the neighbouring unchanged current onto changed elements) instead of zeroing. Removes the root; addresses *both* modes. **Highest leverage.**
2. **Do not over-fresh the opened columns** — blend toward the adjacent open-ocean profile to soften the grid-scale fronts.
3. **Δt ramp** for the mesh-change leg — necessary but *not sufficient* (60 s cleared the fast mode but the slow barotropic one still killed it at day 10); must be combined with (1)/(2).

Everything observed — fronts + black speed panels + w dipoles on the fronts, fast and slow — points to (1): give the changed cells a *balanced* velocity instead of zero.



movcav4_crash_plots/initstate/planview_A_90E_L38.png



movcav4_crash_plots/initstate/planview_D_60s_etacrash.png

2 Current state (2026-07-05, rev. 2, *superseded by §1*): a *time-growing* coastal baroclinic instability on the submesh — *narrowed, not yet root-caused*

Read this first. The OIFS step-0 surface crash that the rest of this document dissects (§8.5 onward) turned out to be a *self-inflicted regression* (a stale, pre-NN-fill ICMGG); with the correctly regenerated ICMGG the OIFS side **works**. The live blocker is that the coupled chunk-3 leg (movcav4, awiesm3 1901) runs ~5–10 model days and then **FESOM blows up** with a vertical-CFL runaway. *This section is the current truth and supersedes §8.5ff.* Today’s work **narrowed the cause substantially**: it is **not** the timestep, **not** the surface flux/ice, and **not** a corrupt or fresh-capped remapped restart. Hourly output plus a targeted diagnostic run point strongly to a **basal-melt shock where the moving geometry puts new ice over above-freezing open-ocean water** (§2.4). A candidate fix is drafted (freezing-point cap on newly-under-ice columns) but has **not yet been tested** in a run.

2.1 Symptom

A vertical-CFL ($\text{CFLz} = w \Delta t / \Delta z$) instability that rings the entire Antarctic ice-shelf margin and grows over days from a clean start. First $\text{CFLz} > 1.75$ warnings appear as early as $\text{mstep} \approx 11$ at ~29 fixed coastal/cavity locations; CFLz creeps $1.8 \rightarrow 2 \rightarrow \dots \rightarrow$ tens over days; eventually a cell’s temperature goes NaN (“found temperture becomes NaN”) and FESOM MPI_ABORTs. The blowup *step* is stochastic (MPI/threading): at **1200 s** it blew at mstep 311–696 (4.3–9.7 d; runs 26060316 seeded / 26055476 unseeded), and the hourly-output run 26061910 at $\text{mstep} = 608$. The *hotspot pattern* is robust and *subsurface* — w and CFLz peak **mid-column at ~95–100 m**, not at the surface. Hotspots (glon/glat, peak CFLz): Totten **108.7/–66.1** (~8), Amery/Prydz 71.4/–70.0, Ross 165.6/–75.0, Weddell/Filchner –44.3/–77.3 & –57.6/–75.9, Bellingshausen –76.5/–72.1, E-Antarctica 93.2/–66.0.

2.2 What we ruled OUT (with evidence)

1. **Anomalous atmosphere heat flux — NO.** Reproduced with OASIS EXPOUT (per-coupling-step dump). The coastal OIFS→ocean heat flux peaks 600–700 W/m² but is **almost entirely solar** (A_Qs_all; Jan solstice at –66°; flux_decomp.py); the non-solar part (A_Qns_oce) is a modest –60 to –120 W/m² (normal open-water loss). Flux *leads*, SST *lags*. Forcing is physically correct.
2. **Missing sea ice ⇒ solar over-absorption — tested, NO EFFECT.** Implemented an ice-donor NN-fill in the F90 remap (§2.6), seeding newly-exposed nodes with the nearest iced neighbour’s ice column (run 26060316, verified 47 455 iced). The CFLz hotspots came back **location-, onset- and magnitude-identical** to the unseeded run. Ice changed *nothing* — the instability is **ocean-internal**.
3. **Timestep / vertical-CFL artifact — revised by the 60 s test (§1): the fast blowup is vertical-CFL; 600 s was simply not small enough.** Halving to **600 s** (step_per_day=144, run 26061443) still blew ($\text{mstep} = 845 \approx 5.9$ d vs the 1200 s baseline’s 4.3 d) with CFLz still 61–71 at depth — which *looked* like “not Δt ”. That inference was wrong: the adjustment w is still supercritical at 600 s, so the runaway regenerates. Going 20× smaller settles it — **at 60 s (step_per_day=1440) the leg makes zero CFLz warnings and runs ≈10 model-days** ($\text{mstep} = 14629$), confirming the *fast* mode is a genuine vertical-CFL instability of the unbalanced initial state. It is **not** a full cure, though: at 60 s a *slow* barotropic eta_n mode instead grows over ~10 days and NaNs (§1). So Δt is a necessary lever for the fast mode but must be paired with fixing the initial imbalance.

4. **Corrupt or fresh-capped remapped restart** — **NO** (and a red herring retracted). The *initial* remapped restart is **clean**: first-wet-level T/S normal (S mean 33.7, no zeros, identical to source), $\text{hnode} \geq 0$ with sane column sums, $\text{ssh} \in [-2.1, 1.2]$ m, $w \approx 1.7$ mm/s at Totten, and **zero NaNs**. *Retraction*: an apparent “2649 nodes with $\text{salt} = 0$ ” is **legitimate under-ice cavity masking** (level-index 0 at cavity nodes with ulevels 2–27), *not* a fresh cap; at each node’s *first wet level* S is normal and matches the source, and *none* of those nodes retreated. The ~ 12 PSU seen in the blowup-file profile is the *evolved* state (the symptom), not the initial condition. **Do not re-chase this**.
5. **Stale OASIS submesh artifacts** — **NO**. `grids.nc`, `masks.nc`, `areas.nc` (feom `srf` = 3.608×10^{14} m²), `rstos.nc` and the `_25` weights are all correct submesh (208 679). The only full-mesh (211 567) leftover was the namcouple feom grid-dimension line — now fixed per-leg (§2.5).

2.3 What the hourly movie shows (the key new evidence)

We added **hourly**, **hourly-split** output of `temp/salt/w` (§2.5) and captured the full onset→blowup. Reading a cross-margin level at ~ 95 m in the Totten region (`movcav4_crash_plots/blowup.TSw_evolution.mp4`; Fig. 3):

- **Hour 1**: clean — smooth T/S , $w \approx 0$ everywhere.
- **~Day 2**: a **fresh + cold tongue nucleates at the retreated cavity margin** near $108.7/-66.1$, with the first up/down w cells beside it.
- **~Day 3.5**: the fresh/cold band **saturates** (< 30 PSU) and **spreads along the whole margin** ($100-118^\circ\text{E}$); the w dipoles intensify.
- **→ runaway**: w grows until the CFLz NaN.

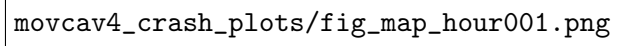
So the blowup is a **coastal baroclinic mode that grows during the run**, at $\sim 95-100$ m, **fed by progressive freshening at depth along the margin** — a *process*, not a static initial-condition pathology (hour 1 is clean; the restart is verified clean).

2.4 rev. 2 hypothesis: a basal-melt shock at newly-under-ice nodes (tested, superseded by §1)

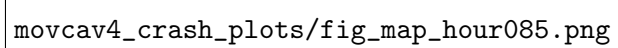
The rev. 2 chain, each step measured (verify-on-output plus a diagnostic re-run with hourly `fw/fh/N2/Kv`, run 26063909): the mesh change puts new ice over above-freezing open-ocean water ($\text{ul}_{\text{new}} > \text{ul}_{\text{old}}$ at **1878** nodes, 667 open→cavity, ringing the margin, drafts to ~ 490 m); the remap copies that warm water straight under the ice; melt fires immediately (`fh` $\sim 250 \rightarrow 2000$ W/m² band on/along the crash front; `fw` ~ 100 m/yr, $100\times$ physical); the freshwater dumped into the ~ 10 m (min 4.6 m) front layer gives $\Delta S = -4$ to -8 PSU/12 h, top-down at the ice base (Fig. 4) → buoyant overturning → CFLz. **This was implemented and tested and does *not* hold as the driver**: the freezing cap `cap_new_cavity_temp` (called at the end of `remap_all_restarts`; drives the front $T - T_f$ to exactly 0 on the 1878 nodes) *and* a full coherent T/S refill were both run in `movcav8`, and **neither prevents the blowup** (§1). Melt is a *secondary* amplifier of a state already unbalanced at $t = 0$ (reconciliation, §1). *Structural aside, still worth a check*: deep-draft open→cavity transitions (130–490 m of ice in one 10-yr PISM step) are physically implausible — verify the submesh `cavity_nlvls` against the PISM ice field.

2.5 Diagnostics + infrastructure that landed (keep these)

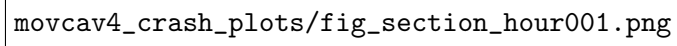
- **Hourly + hourly-split FESOM output** of `temp/salt/w` (the enabler for the movie). `namelist.io` `io_list`: those three set to 1, 'h'; `file_def_fesom.xml`: a new `fesom_1h` file_group with `output_freq="1h"` `split_freq="1h"`. The **hourly split_freq is essential** — XIOS closes+flushes each hour’s file



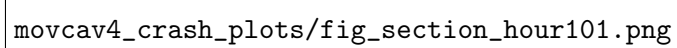
movcav4_crash_plots/fig_map_hour001.png



movcav4_crash_plots/fig_map_hour085.png



movcav4_crash_plots/fig_section_hour001.png



movcav4_crash_plots/fig_section_hour101.png

to disk, so the blowup `MPI.ABORT` cannot lose it (a monthly/yearly split keeps one file open all run and is lost on the crash). Verified: 136 closed hourly files survived the blowup.

- **Movie tooling:** `make_movie.py` (3-panel $T/S/w$ map at ~ 95 m per hour $\rightarrow$ mp4) and `make_section_movie.py` (cross-margin depth-vs-latitude section per hour, under-ice levels masked). `ffmpeg` at `/sw/spack-levante/ffmpeg-`
- **namcouple feom dimension** $\rightarrow$ submesh, per leg: `couple_namcouple` subjob calling `fix_namcouple_feom_dim` in `coupling_ice2fesom.interactive.mesh.functions`; `WORK_DIR.fesom` in `env_fesom.py`. Unblocks OASIS EXPOUT.
- **Ice-donor NN-fill** in the F90 remap (§2.6): correct, kept, but does *not* fix the blowup.
- All plots/movies $\rightarrow$ `/home/a/a270092/esm.tools/movcav4_crash_plots/`, git-ignored (`*.png`, `*.mp4`).

2.6 Reproduce / resume pointers

Experiment `movcav4`, leg `/work/ab0246/a270092/pism_repro/experiments/movcav4/run_awiesm3.19010101-19011231`. Re-run in place with `scripts/movcav4.compute.19010101-19011231.run` (it `srns` directly in `work/`, no `prepcompute` — so edits to `work/` `namelist.*` and `*.xml` take effect as-is). Blowup runs: 26055476 (1200 s unseeded, m696), 26060316 (1200 s seeded, m311), 26061443 (600 s, m845), 26061910 (1200 s + hourly output, m608). **Hourly $T/S/w$ files** are in `work/` (`temp.1h.fesom_*`, `salt.1h`, `w.1h`) — *grab before the next leg overwrites them*. Movie/analysis scripts live in the session scratch; outputs in `movcav4_crash_plots/`. Analysis env: `/home/a/a270092/.conda/envs/pyfesom2_env/bin/python` (base python’s `matplotlib` 3.9 breaks `cmocean`; `imageio-ffmpeg` also installed there). Crash-plot suite: `/work/ab0246/a270092/software/fesom2_crash_analysis` (config points at the `movcav4` submesh + blowup file). Ice-donor code: `couplings/fesom/remap_restart/mo_remap_fields.F90` (thresholds `ICE_CONC_THRESH=0.15`, `ICE_FILL_MAXDIST=150` km; rebuild with `remap_restart/build.sh` — *delete stale `*.mod` first*).

3 Goal and setup

- **Experiment:** `movcav4`, an iterative `esm.tools` coupling. Model 1 = AWI-ESM3 (1-year chunks); Model 2 = Antarctic PISM (10-year chunks). The leg under test is **chunk-3**, AWI-ESM3 year 1901 — the *first* leg after a PISM step has changed the ice geometry.
- **Moving cavity:** PISM ice $\rightarrow$ FESOM sub-mesh (208 679 nodes vs. 211 567 on the full CORE3 mesh). A per-leg tool chain (`ice2fesom`, `ocp-tool`, `remap_restart`) regenerates the OASIS grids/masks/weights (A096 atmosphere grid = 40 320 points, `feom` ocean grid) and the OIFS ICMGG file (land-sea mask, soil type, geopotential, ...).
- **Paths:**
 model `/work/ab0246/a270092/model_codes/awiesm3-develop-is/oifs-48r1`
 experiment `/work/ab0246/a270092/pism_repro/experiments/movcav4`.

4 Coupling-glue performance: per-leg cost profiling and the `fesom2ice` speedup

A separate workstream from the two blockers, recorded here because it governs how fast the iterative campaign can iterate. On each leg the two *models* are cheap (~ 18 min for a FESOM coupled year, ~ 18 min for a 10-yr PISM chunk on 45 nodes), but the *coupling glue* between them dominated the per-leg wall clock. The worst offender was `couple_out_fesom` (the `fesom2ice` ocean $\rightarrow$ ice handoff): ~ 36 min on `movcav5`, run as a single `nproc:1` step while a 45-node allocation sat idle. Same method as the rest of this log — measure, then fix.

4.1 Root cause: 564 redundant whole-file reads, and no interpolation at all

The handoff note blamed a FESOM-unstructured $\rightarrow$ lon/lat *interpolation*. The code refutes that: `couplings/utils/fesom_scalar_array_to_LonLat.py` does *no* horizontal regridding — its output is written on the native FESOM nodes (`mesh.x2/mesh.y2`). The cost was pure redundant I/O. A nested (`time × level`) loop called `pf.get_data(...)` $\approx 12 \times 47 = 564$ times *per variable*, and *every* call re-opened and re-read the entire ~ 262 MB yearly file and recomputed the full time-mean, just to keep one depth level. Worse, because `how="mean"` averages over all times *inside* each call, the outer time loop stored the *identical* annual-mean field into all 12 output slots.

4.2 Fix 1 — read once, vectorise (output bit-identical)

Replaced the nested loop with a single `get_data(depth=None, how="mean")` read plus a vectorised per-level extraction (same `ind_for_depth` mapping, broadcast across time). Validated **bit-for-bit** against the old script (empty `cdo diffn`, identical byte size) and golden-equivalent. Standalone: `temp 283  $\rightarrow$   $\sim 10$  s, salt 537  $\rightarrow$   $\sim 13$  s ( $\sim 24$ – $42 \times$ ).`

4.3 Fix 2 — the whole pipeline ran twice

`iterative_coupling_fesom1x_ice_make_forcing` builds the entire forcing *twice* (a multiyear-mean pass and a total-mean pass); only the final CDO mean differs — the `*_goodgrid.nc` files the second pass rebuilds are byte-identical to the first. Guarded the `temp/salt` build loops to reuse the existing goodgrid when the raw source is unchanged (single-year legs only; rebuilds if the source is newer). *Fixes 1+2 committed: esm-tools 53afb1f3, branch development_interactiv_mesh_awiesm3.*

4.4 Timer instrumentation ([TIMER])

To profile the glue in situ rather than by inference from file mtimes, a lightweight, idempotent lap-timer `couple_timer "label"` was added at the block boundaries of the five runtime-path coupling files (31 marks total: `coupling_fesom2ice.functions`, `coupling_pism2esm.functions`, `coupling_pism2atmosphere.functions`, `coupling_ocean2pism.functions`, `coupling_atmosphere2pism.functions`). After any run,

```
grep '[TIMER]' <exp>/log/*couple*
```

prints a labelled per-step wall-clock breakdown (each mark reports the time since the previous one). Side-effect free, no behavioural change (74d67cc7).

4.5 Ground truth from movcav6, and a corrected picture

The first instrumented leg (`movcav6`) confirms the win *and overturns the assumed bottleneck* (Table 1). Key readings: (i) **Fix 2 works** — `build_grid` pass 2 fell $140.9 \rightarrow 2.6$ s; (ii) `couple_out_fesom` overall dropped ~ 36 min (`movcav5`) $\rightarrow \sim 8.5$ min; (iii) **the real #1 cost is `fesom_ice.select_relevant_variables` (the `cdo monmean`), ~ 155 s/pass — which the handoff note called “sub-second” — and it *also* runs twice (~ 310 s combined);** (iv) `build_grid` in situ (141 s) is $\sim 5.6 \times$ slower than the standalone ~ 25 s, i.e. live-filesystem contention, not the algorithm.

4.6 Fix 3 — the `cdo monmean` was vestigial

The `movcav6` table flagged `fesom_ice.select_relevant_variables` (a `cdo monmean` on each raw 3-D `temp/salt` file) as the #1 cost, ~ 155 s/pass $\times 2$ passes ≈ 310 s. Tracing it shows the `*_var4ice.nc` it produces is **never read**: the Python builder reads the *raw* file directly and does its own time-mean, `concatenate` consumes the `*_goodgrid` files, and `var4ice` is used only to derive the goodgrid *filename* before being deleted. So the `monmean` — a full read+write of the ~ 260 MB yearly file — is pure overhead, left over from when the Python consumed `var4ice`. Replaced with an instant `ln -sf` (goodgrid name and `-f` invariants preserved; 2-D paths unchanged; 08ea580c). This removes

Step (movcav6, real leg)	Wall / s
<i>couple_out_fesom (fesom2ice) — ~507 s total (~8.5 min)</i>	
construct+select (cdo monmean) — pass 1	158.96
build_grid temp+salt — pass 1	140.86
concat+means — pass 1	25.72
construct+select (cdo monmean) — pass 2 (<i>redundant</i>)	154.37
build_grid temp+salt — pass 2 (<i>now skipped, Fix 2</i>)	2.63
concat+means — pass 2	23.60
finalize + write grid/names	1.26
<i>couple_in_pism — ~171 s total</i>	
atmosphere2pism regrid+downscale	19.83
dEBM run (compile+execute)	64.81
ocean2pism cdo remap (per-level weights, 47×)	48.33
ocean2pism extrapolate (fillmiss2)	22.86
(other atmosphere/ocean steps)	~15
<i>couple_out_pism (pism2esm) — ~119 s total</i>	
get_newest_output (cat+timmean, 120 steps)	88.58
ice-geometry seltimestep/expr	24.14
(orography/thickness anomalies, runoff, assemble)	~6

Table 1: movcav6 per-step coupling-glue timings from the [TIMER] instrumentation (grep '[TIMER]' on the couple_* logs).

both passes' monmean and takes couple_out_fesom from ~508 s toward ~200 s — i.e. the fesom2ice handoff is now ~36 min → ~3–4 min across Fixes 1–3.

4.7 Remaining ranked targets

1. pism2atmosphere get_newest_output (cat+timmean, ~89 s) — largely *inherent*: 10 yr of monthly PISM extra output (120 steps, 2.36×10^9 values). Low-yield/high-risk.
2. ocean2pism cdo remap (~48 s) — distance-weighted weights are regenerated *per depth level* (47×) because the wet/dry source mask varies by level; the source geometry is fixed, so the weights could be built once.
3. Minor: pism2esm ice-geometry seltimestep (~24 s); ocean2pism fillmiss2 (~23 s); a fixed sleep 10 at the end of ocean2pism.
4. *Not* a target: couple_in dEBM (~65 s) — the “compile” is a fast cp \$DEBM_EXE; the time is the dEBM Fortran model *executing* on a 1.1 GB input, i.e. genuine compute.

Net: the fesom2ice handoff is ~36 min → ~3–4 min (Fixes 1–3). The remaining PISM-side glue (~8 min, mostly inherent large-file CDO) is the next frontier if further speedup is wanted.

Part II — The OIFS step-0 coastline crash (resolved; historical record)

*Everything from here to §13 documents the first blocker, now **solved**. It is kept as a full record of how the OIFS coastline crash was isolated and fixed; §2 supersedes it as the live state of the coupling.*

5 Infrastructure fixes already landed (coupling foundation)

Before the surface crash could be isolated, a series of framework/data bugs had to be cleared (all now resolved and validated at run time); these underpin *both* blockers:

- **OASIS weights/restarts:** fresh 208 679-point weights and remapped OASIS restarts are staged in the correct sequence; the MCT sparse-matrix dimension mismatch is gone. `rstos` SST is valid (271–305 K, no zeros); the `feom` → A096 SST remap is exact (all 28 528 ocean cells row-sum = 1.0).
- **ICMGG regeneration:** the regenerated ICMGG land–sea mask equals the A096 OASIS mask *exactly* (11 792 land points, zero disagreement).
- **esm_tools guards:** an `observe.py` race (compute launching before `couple.in` finished), `remap_oasis_restart` sanity guards, a shared `rmp-pool` write guard, and per-leg `plit` generation/wiring.

Conclusion of this phase: every component we *regenerate* agrees on the new coastline, so at that point the remaining crash was isolated to the OIFS side at initialisation — pursued through the rest of Part II.

6 Symptom: a moving-target step-0 crash

The leg dies during step 0, always on the same handful of MPI ranks — the ones holding the Antarctic coastline. As each cause was removed the crash moved *downstream*, which is itself the main diagnostic signal:

Build	Crash	Where / meaning
baseline	error (73) divide-by-zero	radaca: ρ , PTH/PTH built from skin $T = 0$.
+skin-T reseed	error (73) again	skin T re-zeroed: ocean scheme sets skin $T = \text{SST}$, and SST was 0.
+SST reseed	error (65) floating-invalid	advances past the whole surface/radiation chain; now NaN in <code>cpdyddh:915</code> (DDH moist-physics budget, $\text{PQR} \times \Delta p$).

So the surface divide-by-zero is *solved*; the live failure is a NaN precipitation flux produced by the moist physics on the same cells. (§8.5 later revised the *attribution* of the first two rows: the reseed acts only on `NPROMA` padding, and it is Approach A’s ICMGG re-ingest that actually populates the real flipped cells — but the observed crash-migration downstream is real either way.)

7 Method: measurement over speculation

Each of the findings below comes from an instrumented build (a debug `WRITE` at a specific line), run for five minutes on compute nodes, not from reading the code alone. The instrumentation lives behind the `ECE_LANDICE` path so it does not touch ordinary runs.

8 What the measurements established

8.1 Skin temperature alone cannot be seeded

A step-0 reseed in `upatim.F90` that set the skin-T reservoir to the freezing point `RTT` had *no effect*: the surface scheme re-derives ocean skin temperature from the SST later in the step, so with

SST = 0 the seed is overwritten back to zero before the radiation call. The SST field itself has to be seeded.

8.2 Seeding SST appears to clear the divide-by-zero

Adding an SST reseed ($SD_VF\%YSST \leftarrow RTT$ where $< 100\text{ K}$) coincided with the error (73) giving way to a downstream failure. Instrumentation showed no cell degenerate by the end of UPDTIM (DBG-UPDTIMEND: 0 cells still $skinT(MP9) < 100$), and the model advanced into $cpg_drv \rightarrow cpg \rightarrow cpg_dia$ before failing at

```
forrtl: error (65): floating invalid
  cpdyddh_ 915  cpdyddh.F90      ! PDHCV(..)=PQR(JROF,JLEV)*ZDELPREV(..)
```

The invalid operand is a rain/snow flux PQR/PQS out of the moist physics — a genuine physics NaN, not a diagnostic artefact. (§8.5 shows the reseed was actually a no-op here; the real cells were already valid because of Approach A. The cpdyddh NaN is the true remaining failure regardless.)

8.3 The coupler delivers a valid SST to every ocean cell

The obvious next hypothesis was “the newly-wet cavity cells receive no SST from FESOM”. This was tested directly: in `ece-updc1ie_cpl.F90`, at the exact line that applies the coupled SST to ocean-gated cells,

```
IF (SD_VF(JROF,YLSM%MP,IBL) <= 0.5 .AND. SD_VF(JROF,YCLK%MP,IBL) <= 0.5) THEN
  SD_VF(JROF,YSST%MP,IBL) = CPL_FLD_SST(IGP)      ! <-- incoming FESOM SST
```

we counted, per rank, the ocean-gated cells whose *incoming* CPL_FLD_SST is degenerate ($< 100\text{ K}$), printing each cell’s longitude, latitude, masks and `plit`. Result, run 26030286:

```
DBG-CPLSST-TOTAL: 0 ocean-gated cells got coupled SST<100 K, of NGPTOT=105    (x384 ranks
)
```

Zero on all 384 ranks. The coupler supplies a valid SST to every ocean cell. The “FESOM is not delivering SST” explanation is therefore **refuted**, and with it the idea that the fix lives on the ocean/OASIS side.

8.4 The “degenerate cells” were NPROMA padding

A per-cell classifier (DBG-FLIP) in `updtim.F90` dumped, for every cell with a degenerate skin T or SST *before* any reseed, its `lon`, `lat`, `LSM`, `CLK`, `CI`, `SST`, `skin T` and `plit` — but guarded on the real-grid bound $IGP \leq NGPTOT$. The result was decisive:

```
DBG-FLIP-TOTAL: 0 real cells degenerate (skinT<100 OR SST<100) before reseed    (x384)
```

Zero real cells. The “~7 degenerate cells per rank” that the reseed counter had been reporting are the NPROMA *padding* slots beyond NGPTOT (which are legitimately zero); the reseed’s `DO JROF=1,NPROMA` loop counted them, the classifier’s `NGPTOT` guard excluded them. So the reseed is a no-op on the real grid, and the moist-physics NaN is *not* caused by a degenerate skin T /SST on any real cell. This reversed the working picture and forced the next question: what actually differs on the crash cells?

8.5 The crash cells are exactly the cells re-ingested from ICMGG

The answer was found by realising the tree already contains an implementation of *Approach A* (§10): `reressf_part2.F90`, gated on `ECE.LANDICE`, reads the regenerated ICMGG with `SUGRIDG` on a warm start and re-applies its surface column — via a new masked `PUTALLFLDS_M` operator in

`surface_fields_mix.F90` — **only at grid points whose land–sea type flipped** ($|\text{LSM}_{\text{icm}} - \text{LSM}_{\text{srf}}| > 0.5$). Instrumenting it (DBG-REINGEST, run 26032081) shows it is *surgical and physically sane*:

```
DBG-REINGEST igp=41 lon=58.97 latsin=-0.919 LSMsrf=6.29E-2 LSMicm=1.00 sktICM=270.4
  stl1ICM=270.3
DBG-REINGEST igp=63 lon= 0.00 latsin=-0.938 LSMsrf=1.00    LSMicm=0.37 sktICM=271.8
  stl1ICM=271.8
... total 70 cells across 16 ranks:  49 sea->land,  21 land->sea
```

70 cells total — the actual coastline change — with injected skin/soil temperatures of 262–272 K (all physical, no zeros). And the ranks it touches (2155–2174) are *exactly* the ranks that then NaN in `cpdyddh`. This is why §8.3 and the padding finding both came back clean: the re-ingest had already given every real cell a valid surface column. The problem is not the surface *state* — it is that flipping a cell’s land–sea *class* at run time, most acutely the 49 sea → land cells where a warm moist marine atmospheric column is now over frozen land, is what the first moist-physics step turns into a NaN.

Arbiter: the re-ingest is necessary, not harmful. A build with the re-ingest *disabled* (LLREINGEST_DBG=.FALSE.) does *not* run further — it reverts to the *original* error (73) divide-by-zero in `radaca` (skin $T = 0$). With the re-ingest enabled the model instead advances *past* the whole surface/radiation chain and dies one subsystem later in `cpdyddh`. So Approach A is doing real work: it repairs the surface so the flipped cells are no longer degenerate. What remained was a *second*, independent problem — pursued in §11 and §12.

9 Background: why the runtime mask can disagree with ICMGG

Independently of the crash mechanism, it is worth recording that on a warm start the OIFS runtime land–sea mask does *not* come from the regenerated ICMGG at all. The restart path (`cnt3_glo.F90`) is:

```
IF (NCONF == 1 .OR. NCONF == 302) THEN
  CALL RERESF_PART1 (...)    ! sets LNF from restart-control file: present -> LNF=.FALSE
  .
  CALL RERESF_PART2 (...)    ! srf bulk restore (GPOPER 'PUTALLFLDS'): ALL slots incl.
  LSM
ELSE
  LNF = .TRUE.
ENDIF
IF (LNF) CALL CSTA(...)      ! ICMGG read (SUINIF/SUGRIDF/SUGRIDG); ONLY on a cold start
```

`PUTALLFLDS` (`surface_fields_mix.F90`) copies every NDIM storage slot with no per-field flag, so the land–sea mask is always overwritten from `srf`; and because a `srf` restart is present each leg, `LNF=.FALSE.` and `CSTA` — the only reader of the regenerated ICMGG geometry — never runs. A KTRAJ tagging attempt in `su_surf_fds.F90` was ruled out: KTRAJ controls the trajectory/FullPos write, not the bulk restore, so it had no effect on the runtime mask. This stock behaviour — a stale, `srf`-restored mask on every warm start — is precisely what Approach A (§10, now implemented in `RERESF_PART2`) was written to override at the flipped cells.

10 The two approaches, and where they now stand

Two designs can make OIFS respect the moving coastline. §8.5 showed the first is *already built and running*; the open question is no longer “which to implement” but “is the implemented one physically viable, or must we fall back”.

Approach A — re-ingest the regenerated ICMGG geometry on a warm start (*implemented, live*). `reressf_part2.F90` (gated on `ECE.LANDICE`) reads the regenerated ICMGG via `SUGRIDG` and, using the new masked `PUTALLFLDS_M` operator (`surface_fields_mix.F90`), replaces the *entire* surface column at the 70 cells whose land–sea type flipped, leaving every other cell on its spun-up `srf` state. §8.5 confirmed it does exactly this, with sane values. *The concern it raises*: it changes a cell’s land–sea class between two model states in a single step. The atmospheric column above a sea → land cell is still a marine column; the surface beneath it is now frozen land. The first moist-physics step over that mismatch is the current NaN suspect. If the disabled-re-ingest run comes back clean, this is confirmed and Approach A needs either a gentler transition (e.g. only flip where `plit` also marks ice, so the cell enters the *glacier* path rather than bare-soil land) or a matching adjustment of the near-surface atmosphere.

Approach B — the established `plit` overlay. EC-Earth’s ice-sheet coupling (documented in `ece4-pism-coupling.md`) does *not* re-read or flip the LSM. It keeps the `srf` geometry and communicates the evolving ice as a fractional mask `plit`: `suorog.F90` patches the orography from `usurf_oifs` and (with `ECE.LANDICE`) reads `plit_oifs` at leg init, and the live `PLIT_<region> OASIS` field drives the *glacier* physics via `SURFACE.GET LANDICE`. *Pros*: it is the design the OIFS ice code was written for, and it never creates the abrupt land–sea-class flip that Approach A does. *Cons*: it depends on `plit_oifs` being generated (`make_plt_oifs.py`), read and applied, and on the live `PLIT` path (the ISM-mapper OASIS component), which the `movcav4` setup may not yet run.

Where the evidence points. Approach A is confirmed necessary (§8.5), and it is kept. But it faithfully transfers whatever the coupling tool chain writes into the ICMGG — so if a flipped cell is inconsistent there, the re-ingest dutifully installs that inconsistency at run time. That turned out to be the actual bug (§11), and it lives in the tool world, not in OIFS.

11 The real root cause: `ocp-tool` leaves flipped cells half-converted

An FPE-safe operand check at the crash line (`DBG-DDH`) showed the non-finite operand is `PCOVPTOT` (the precipitation-cover fraction from the moist physics), not the pressure thickness. Following that upstream to the surface state of the crashing cell (a sea → land cell at 74.3°S, 257°E) exposed the flaw — in the regenerated ICMGG it reads

```
lsm=1.00  slt=6  swvl1=0.000  stl1=272.0  ci=0.838  sst=270.6  <-- half ocean, half
land
```

i.e. a *land* mask over an otherwise *ocean* column: zero soil moisture, a leftover sea-ice fraction and an SST. `ocp-tool`’s `lsm.py` (`modify_lsm`) flips a cell by setting *only* the land–sea mask and the soil type (`slt=6` = “sandy clay loam”), and leaves *every other surface field* at its old-type value. The moist physics over that mongrel column manufactures the NaN.

Fix (tool world, committable). `modify_lsm` now rebuilds the *whole* surface column of each flipped cell from the nearest stable neighbour of its *new* type (a masked great-circle nearest-neighbour fill, `fill_flipped_from_nearest_neighbour`). A newly-grounded cell then looks like the ice/land around it (`slt=1`, `swvl1`≈0.30, `ci` = 0, glacier snow), a newly-wet cell like the ocean around it — with the freshly set land–sea mask preserved. Regenerating this leg’s ICMGG with the fix turns the crash cell from the mongrel column above into `lsm=1`, `slt=1`, `swvl1=0.302`, `ci=0.000` — a self-consistent glacier-land column. This is the right home for the fix: OIFS stays generic and the coupling machinery owns coastline consistency.

12 The residual cpdyddh NaN was DDH-diagnostic-only

With the consistent ICMGG, the model still stopped at cpdyddh — but that routine is a *DDH budget diagnostic* (Diagnostics of Horizontal Domains), and the crash line accumulates a *passive* precip-cover variable. Disabling DDH in the namelist (NAMDDH: LHDGLB=LHDPRG=LHDHKS=.false.) removed it, and OIFS then ran **~700 coupling steps** without any FPE trap. Since OIFS is compiled -fpe0 (every arpifs object carries -fpe0 -fp-model precise), running that far proves its forecast state — and the fluxes it sends the ocean — are finite: the PCOVPTOT NaN was confined to the DDH passive-variable accumulation (an uninitialised diagnostic slot at step 0), not the real physics. OIFS is no longer the blocker.

13 OIFS crash — final status (resolved)

Item	Status
Coupling machinery (mesh, OASIS, ICMGG regen)	Consistent & validated; ICMGG mask = A096 mask exactly; ocean SST delivery 0/384 degenerate.
OIFS surface divide-by-zero	Solved by the ICMGG re-ingest (Approach A, RERESF_PART2), confirmed necessary by the disable arbiter.
Flipped-cell inconsistency (ocp-tool)	Fixed — masked nearest-neighbour rebuild of each flipped cell’s surface column (§11); verified in the regenerated ICMGG.
cpdyddh NaN	Diagnostic-only; disabling DDH (NAMDDH) let OIFS run ~700 coupling steps under -fpe0. Real physics/fluxes are finite (§12).
OIFS overall	No longer the blocker.
New frontier: FESOM	Superseded by §2 (the current truth). This row’s open question — “finite-but-extreme OIFS forcing” — has since been ruled out (§2): the blowup is a <i>time-growing coastal baroclinic instability</i> on the submesh, ocean-internal, driven by subsurface freshening at the retreated margin; timestep, flux, ice and restart corruption are all excluded.

Key files: ocp-tool/lsm.py (modify_lsm + fill_flipped_from_nearest_neighbour — the tool-world fix), arpifs/control/reresf_part2.F90 + module/surface_fields_mix.F90 (re-ingest PUTALLFLDS_M), arpifs/dia/cpdyddh.F90 (DDH crash site), NAMDDH in fort.4 (DDH off). Debug instrumentation (DBG-*) in updtim.F90, ece_updclie_cpl.F90, climaer_layer.F90, cpdyddh.F90 is to be stripped before committing OIFS.

14 To do

- ☐ ocp-tool NN fix — *done*: committed + pushed, PR #51 (JanStreffing/ocp-tool, branch fix/flipped-cell-nn-fill → master).
- ☐ **Commit the OIFS re-ingest** (branch movcav-landice, SMHI fork). First clean it up:
 - *keep*: surface_fields_mix.F90 (PUTALLFLDS_M, no debug), bundle.yml;
 - *strip*: the DBG-REINGEST block and the LLREINGEST_DBG toggle in reresf_part2.F90 — make the re-ingest unconditional under ECE_LANDICE;
 - *revert entirely* (pure debug / no-op): updtim.F90 (reseed + DBG-FLIP), ece_updclie_cpl.F90 (DBG-CPLSST), climaer_layer.F90 (DBG-CLIMAER), cpdyddh.F90 (DBG-DDH).

- **Make DDH-off permanent** in `esm_tools` — fold the `NAMDDH LHDGLB=LHDPRG=LHDHKS=.false.` into the OIFS namelist config (likely `configs/components/oifs/oifs48.cmip.yaml`) rather than the runtime `fort.4`.
- **Curate + commit the `esm_tools` WIP** (branch `development_interactiv_mesh_awiesm3`): the OASIS-regen/plit wiring in `coupling_ice2fesom_interactive_mesh.functions`, the `remap_oasis_restart.py` guard, the `observe.py` race fix, the `awiesm3.yaml` pool guard — split into logical commits.
- **Root-cause the live FESOM instability (§2)**. The frontier is no longer OIFS: it is a time-growing coastal baroclinic mode fed by margin freshening. Next probes, in order: (a) the sub-shelf **meltwater/freshwater flux** on the submesh where the cavity was just exposed; (b) submesh **element quality** / `zbar-hnode` **consistency** at the new coastline; (c) whether the freshening is surface meltwater mixed down or a distributed sub-shelf source (section movie done, §2.3).